

On the restriction to numeric indicators in Performance Measurement Systems

Tim Pidun and Carsten Felden

Chair of Information Science
TU Bergakademie Freiberg
Freiberg, Germany

tim.pidun@bwl.tu-freiberg.de, carsten.felden@bwl.tu-freiberg.de

Abstract—Many business processes cannot be easily measured due to their non-deterministic or qualitative nature. So, to fit in common performance measurement systems (PMS), artificial and simplifying measures are used that are complicated and costly to create and evaluate. Using a literature review, this paper documents the dominance of PMS that rely on KPIs and a lack of those that also incorporate non-numeric, generic indicators that better address qualitative problems. Considering the need for better transparency and comparability of business processes as well as visibility of process performance, the final necessity of deployment for a refined PMS using additional indirect indicators is derived. It would be able to assess hidden performance problems and to reveal additional improvement possibilities.

Keywords- *BPM, KPI, Indicator, Process Performance Management, Performance Measurement System, Visibility, Measurement paradigm*

I. INTRODUCTION

Numeric Key Performance Indicators (KPIs) evaluate business processes and allow the tactical management of an enterprise to judge if value will be delivered [1]. Measurement is the carrier to monitor and control according to the Business Process Management (BPM) approach (e.g. Bucher and Winter [49]). The restriction to quantify performance in so-called performance measurement systems (PMS) makes it difficult to assess hard-measurable or business processes that are rather qualitative in nature and better could be described by e.g. success factors, soft goals, milestones, complexity, relations, dependencies or maturity.

Recent research of Raschke and Ingraham [2] indicates strong evidence that for example a high business process *maturity* in key processes of the production environment positively affects business performance and hence lead times, inventory and holding costs which ultimately reduce cost of goods sold and increase the gross margin. Anderson et al. [47] as well as Jeanes [48] also found a significant positive correlation between *customer satisfaction* and *financial* performance. Additional improvements that can be read out by other additional indirect indicators additional to KPIs are likely to contribute to process success and outcome as well and hence, to contribute to business performance.

Following example outlines the problem area.

In a project that was outlined to assess the *administration* processes in a German university, the originally intended assessment method was *Six Sigma*, which is build to ensure reproducible quality of a process by a feedback cycle of definition, measurement, analysis and control [46]. But since processes in public administration and in particular in higher education, e.g. *exam management* (1) are rather of qualitative nature and so, are not easily assessable by numeric values like time, cost or productivity, (2) the statistically reproducible *success* of the process is not questioned, but (3) rather the outcome in the sense of effectiveness of the process was in focus, the use of numeric values for process assessment in the sense of building and controlling *common* KPIs was hardly possible. As we were looking for an alternative PMS that could handle this problem, we were always put back to the attempt to *abstract* qualitative cases to numeric values or add more and more *viewpoints* and hence, measures, which in this case wasn't helpful. We expected a system that uses more *indirect* values to describe our problems, like amount of steps and elements or media interruptions, a probability of rework, complexity or – if numeric values weren't useful at all, roles, responsibilities and their relations as well as the possibility to use certain goals as indicators of success, but didn't find such a system. In other words, we would have had advantages by using a system incorporating *indirect* and *non-numeric* indicators because we would have been able to detect performance that was *invisible* to KPIs.

So, from a broader view it is the paper's goal to emphasize the drawbacks in business performance measurement in case of the restriction to the use of KPIs in PMS.

There are three main reasons to expand the scope of the assessment of business processes to additional indirect indicators:

- (1) Out of business needs, an enterprise should be able to evaluate their processes' efficiency and to rate their effectiveness in any given business process without restriction to a specific process *domain*, performance system or measure constraint, which would be the case when only using numeric KPIs [3]. It should also not be forced to use many *isolated* models for the

- analysis and assessment of its processes to overcome this restriction.
- (2) The evaluation of relevant IS theories according to this context (e.g. Porter's five forces, the principal-agent-theory and the organizational information processing theory) led to *additional requirements* that a business and hence its business processes need to become more *transparent* and *comparable* irrespective of their *functional domains* in order to be assessable and hence optimizable. Subsequently, results, constraints and properties of processes would be more visible [ibid].
 - (3) To enhance *visibility* of process performance. The concept as the principal ability to assess process performance information [50] is not restricted to numeric parameters, but considers all main types of information, the most important features of the information flows and an appropriate set of metrics. It is established by using a performance assessment system that gives appropriate information through numeric and verbal performance descriptions.

The contribution of this paper is the proof that state-of-the-art PMS are not able to fulfill these three additional requirements and business needs. In contrary, a need for a PMS framework solution which is able to reveal additional performance potential is derived, by integrating additional non-numeric concepts for assessing business processes.

Proceeding with the description of the state of the art, then the used research design is proposed. In the investigation course part, the discussion of the PMS and complete concepts follows. Finally, the conclusion and outlook part frame this paper.

II. STATE OF THE ART

In this state of the art section, the theoretical grounding of measurement, currently used concepts of PMS approaches and complete concepts are discussed. Out of these examinations, three different viewpoints to the state of the art are derived. Finally, the academic void is being postulated by raising research questions and hypotheses which are the basis to the Investigation course.

A. The measurement paradigm

The work of Power [51] supports the understanding why measurement is a *must have* in enterprises in principle. He claims the existence of an audit society that gave up the concepts of trust and confidence in favor of compliance and assurance. Auditing mechanisms are implemented by delegation of control and the use of certification systems in control chains, in which neighboring elements supervise each other as well as become accountable themselves (controlling of control). As Zerfass [52] states, this paradigm is visible throughout all parts of society, which is the reason why so many parts of life are considered to become measurable nowadays more and more, even those that weren't in the focus of auditing and assessment before, e.g.

non-profit organizations [29] or education [53]. The same is true for business domains.

More and more processes are expected to become measurable though the qualitative nature of their results or outcome and though they do not necessarily feature a direct causal dependency of input and output. The assessment of these control chain elements has to be done via the assumption of simple interdependencies and by the use of simplistic measures in order to feed the audits. Hence, control chains are established in order to make the processes controllable in a supposedly more intersubjective way [52]. This leads to the danger of reducing assessment to rituals of verification [51], which means number crunching to display measures in audits instead of other meaningful assessment methods.

In fact, business domains like e.g. PR [52] or human resources [54] already are subject to control by simplistic numeric values in common PMS in order to feed the control chains. In business, numeric values are called KPIs and are the data source for a PMS, which embodies an element of control in Power's sense to control business processes.

B. Performance Measurement Systems

The definition and perception of what performance measurement should stand for is not cohesive in the literature, and the subject of research has to be better specified. The terms *performance* and *success* in combination with measurement are often used synonymously in most publications and have to be reviewed in detail. Most authors agree on the need for a performance measurement framework in principle (e.g. [4], [5]), but the scope remains different. Neely [6], Bourne [7] and Miles [8] prefer the definition of performance measurement as a process of quantifying action rather than building a framework for future systems. Similarly, Lönnqvist [9] suggests defining a performance measurement system as a set of measures used to determine the status of attributes of the measurement objects.

In an exemplary approach, DeToni and Tonchia [10] try to classify existing PMS into models that are strictly hierarchical, Balanced Scorecard (BSC) models, frustum models (aggregated measures without link to financial performance), models combining internal and external performances and value chain related models. Pun and White [11] divide them by their purpose into self-assessment and improvement facilitation, and Franco-Santos [12] even exclude public and no-profit sector literatures in their research for a definition of business performance measurement (BPM) systems and still count *seventeen* different definitions.

C. Complete concepts

Despite of critical discussions concerning the possible restriction in the meaning of the term *measurement* to numeric indicators (e.g. Lönnqvist [9] as well as Pidun and Felden [3] propose *assessment* as more adequate to also include other indicators), the term performance measurement system (PMS) is accepted as common wording of the artefacts in discussion. They normally refer to frameworks

that describe *how* a system is performing, not to models defining or offering an approach for directly *analyzing* them.

Neely [6] in particular differentiates performance assessment as process of quantifying the effectiveness and efficiency of an action and performance measurement being the metric used to quantify the effectiveness and efficiency of an action, either in terms of the actual status or the end result of the action.

We synonymously use the terms effectiveness in this paper to describe to what extent customer's requirements are met at all (outcome) and efficiency to describe how economically the firm works with their given resources (success). This definition is similar to

- (1) Drucker [13] who describes effectiveness as the outside (customer) contribution of a manager's efforts; them being assessed by results, as well as
- (2) a common distinction between doing the right things (effectiveness) and doing things right (efficiency).

KPIs primarily measure process *performance* (i.e. the success of a process) whereas a certain process *quality* is not sufficiently addressed - by means of its capability to meet the customer's requirements under all possible constraints [14].

Reviewing the literature on PMS, there are visible differences in their viewpoints between design recommendations for a performance measurement system

- (1) existing performance measurement systems and their constraints and
- (2) descriptions of the measures/indicators attached to them.

We propose to spotlight findings and problems grouped to this nested set of viewpoints. They are discussed in detail in the investigation course section.

D. Research gap

The main question out of these observations is why quantizing and the exclusive use of numeric indicators should be a necessity, or, more provoking, seems to be a dogma in PMS that has never been questioned yet.

It might be hard to find, to calculate and control numeric measures for qualitative performance problems that only can be quantized insufficiently, like customer satisfaction, milestone achievement, inherent complexity or maturity. Though, numeric indicators seem to remain the main and in most cases, the *only* indicators in today's PMS. So we raise following research questions:

- Can principal limitations of PMS be found and articulated throughout the literature?
- Can the dominance of PMS using KPIs be proved?
- Are KPIs able to reveal performance potential in processes?
- To what extent can PMS fulfill the additional requirements for business processes?
- To what extent can PMS ensure a good visibility of performance?

- How could a PMS solution look like that overcomes these disadvantages?

III. INVESTIGATION COURSE

This section consists of five parts explaining the research design of this contribution: First, the empirical research through a literature review, second, the reflexion about the three different viewpoints to the PMS introduced in Section 2 and finally the findings, in which part the research questions are processed by matching the found properties, functions and limitations of PMS to the principles to make business processes more *comparable* and *transparent* that result from the discussion and application of relevant IS theories (additional requirements).

A. Literature review

To gather information on the matter, we decided to perform a literature review. It was carried out from January to November 2010 based on 70 sources discussing performance measurement approaches, their application, advantages, problems or distinctive features out of online and printed journals, books and practitioner reports. They were reviewed using the systematic review scheme according to Denyer and Tranfield [15].

Used databases were Emerald, Springerlink, EBSCO Source Complete and Google Scholar, including reverse citations where possible. Literature that compared and reviewed PMSs on their part could also be identified and was taken into account for a meta-review.

B. Recommendations for PMS' design

There can be different approaches of building a PMS from design to review. Wisner and Fawcett [16] offered a nine-step rulebook: define the firm's mission statement, identify strategic objectives, clarify functional area roles, develop performance measures, communicate strategic objectives and performance goals, assure consistency with strategy, assure the compatibility, use the performance measurement system and periodically re-evaluate. Medori and Steeple [17] propose another design approach for auditing and enhancing PMS in six stages: define success factors as well as the performance measurement grid of quality, cost, flexibility, time, delivery and future growth, select measures, audit, implement, and maintain. Their framework also contains 105 measures.

Machland and Raymond [18] offer three criteria that a PMS should focus on: management support sophistication, IT sophistication as well as alignment and scope. Smullen [19] counts five basic requirements to a PMS: acceptable, suitable, feasible, effective and aligned between non-financial measures and financial goals.

DeToni and Tonchia [10] remember to integrate a PMS into the enterprises' ecosystem of accounting system, manufacturing planning/control system and the strategic planning. Finally, Machland and Raymond [18] state that multidimensional, balanced or integrated models surely can be developed from PMSs including various types of

indicators, as well as being managed in a coordinated way. They do not mention a *restriction* to numeric indicators in principle.

C. Obstacles to PMS' implementation

The decision to apply a specific PMS in an enterprise does not necessarily mean successful implementation. Some compilations on key obstacles to PMS implementation exist; Bourne [7] e.g. postulate in their work two main drivers of success (benefit from the project and top management commitment) as well as four main barriers (effort required, data accessibility, consequences of measuring and disruption by new initiatives). Kaplan and Norton [20] go in a slightly different strategic direction when claiming their four found obstacles as: vision and strategy non actionable, strategy not being linked to department, team or individual goals, the strategy being not linked to resource allocation as well as the feedback being tactical and not strategic.

More detailed information about significance and severity can be extracted by clustering the particular reasons for bad implementation throughout the regarded literature:

- (1) The problem of the top management no longer being focused or *committed* to the PMS is been seen as one of the heaviest roadblocks by Bourne [21] and Franco-Santos and Bourne [4]. In contrary, striving for *perfection* is also been seen as obstacle going in the opposite direction in Bourne [22].
- (2) The unreflected usage of operational targets and goals *unlinked* to strategy, culture, or departments or diverging with time is mentioned in Kueng and Krahn [23], Bourne [21], Chennell [1], Ittner and Larcker [24], Franco-Santos and Bourne [4] and Franco-Santos [12]. Franco-Santos and Bourne [4] also mention a certain lack of refinement and *review* loops. Especially the used *measures* are subject to critical review in the literature.
- (3) The problem of unadapted *relying* on generic frameworks or measures that are poorly defined and might not actually contributing to the specific economic results is discussed in Bierbusse and Siersfeld [25], Schneiderman [26], Bourne [22] as well as Ittner and Larcker [24].
- (4) Having a large number of measures *diluting* the overall impact is being seen an obstacle by Brown [27], Bierbusse and Siersfeld [25] and Bourne [22].
- (5) *Resistance* to measurement and change are seen as key blocking factor by Bourne [7], [22] as well as Franco-Santos and Bourne [4].
- (6) Meekings [28] and Brown [27] amend lack of understanding and *fear* of personal risk or competition to be important obstacles.
- (7) Bierbusse and Siersfeld [25], Bourne [21] and Lönnqvist [9] remember that having computer system issues while the implementation of a PMS information system is of course a key aspect, too.
- (8) Bierbusse and Siersfeld [25], Kueng and Krahn [23] as well as Lönnqvist [9] mention the fact that

redesigning or gathering data would have generated extra *cost* not being linked to the potential value (of a parameter that should indicate economic efficiency).

Finally, one of the most important obstacles is

- (9) The need or unwillingness to quantify results in areas that are more qualitative in nature, intangible or hard-measurable (Bierbusse and Siersfeld [25], Kueng and Krahn [23], Bourne [21], Ittner and Larcker [24], Lönnqvist [9], Moxham [29] and Rehage [30]).

D. Restriction to numeric indicators

By far most of the reviewed PMS are *exclusively* using numeric measures as indicator of performance, success or value and are considering measures (also: metrics or data) as necessary or sufficient requirement for a performance system. Stalk and Hout [31] explicitly demand the measure to be *quantitative* (and close to the customer).

Though Bourne [21] claim that the implementation of a measure does not at all mean to have created a performance measurement system and Tangen [32] remembers to remove old performance measures that are no longer of interest to the business performance system, all in all the basic concept of applying only a numeric measure is *not* questioned.

Some collected examples of Performance Measurement Systems using numeric indicators are, sorted by their relevance and distribution in practice and literature:

- Balanced Scorecard (BSC) by Kaplan and Norton [20], discussed amongst others by Ghalayini [33], Kueng and Krahn [23], Bourne [7], [22], Lin [34], Ittner and Larcker [24], Beatham [35], Lönnqvist [9], Tangen [32], Craciun and Criveanu [36], Folan and Browne [37], Garengo [38], Pun and White [11] as well as Neely [6].
- Performance Prism by Neely [39], discussed by Bourne [22], Ittner and Larcker [24], Lönnqvist [9], Tangen [32], Folan and Browne [37], Garengo [38] and Pun and White [11].
- SMART/Performance Pyramid by Lynch and Cross [40], discussed by Ghalayini [33], Bourne [22], Tangen [32], Folan and Browne [37], Garengo [38] as well as Pun and White [11].
- Results/Determinants Matrix by Fitzgerald and Moon [41], discussed by Lin [34], Bourne [22], Folan and Browne [37] and Garengo [38].
- Performance Management Questionnaire by Dixon [42], discussed by Neely [6], Ghalayini [33] and Bourne [21]
- Supportive Performance Measures/Performance Measurement Matrix by Keegan [43], discussed by Neely [6], Bourne [22] as well as Folan and Browne [37].
- Sink and Tuttle Model by Sink and Tuttle [44], discussed by Tangen [32] as well as Folan and Browne [37].
- Dynamic Performance Measurement System by Latintnen [45], discussed by Lönnqvist [9].

- Skandia's Intellectual Capital Navigator by Edvinsson [46], discussed by Ittner and Larcker [24].
- Integrated Dynamic Performance Measurement system by Ghalayini [33], discussed by Pun and White [11].
- CMMI by Kneuper [47], in this context discussed in a previous version by Kueng and Krahn [23].
- Integrated Performance Measurement System by Bititci [48], discussed by Pun and White [11].
- Ambite Performance Measurement Cube by Bradley [49], discussed by Folan and Browne [37].
- Integrated Performance Measurement Framework by Rouse and Putteril [50], discussed by Folan and Browne [37].
- Intangible Success Factors, introduced by Lönnqvist [9].
- Brown's Process-oriented Framework, introduced by Brown [27].
- Organizational Performance Measurement, introduced by Chennell [1].

Some other approaches could be found that are named PMS, but after a detailed text drilldown turn out to be assessment methods and tools for evaluating measures rather than frameworks:

- Neely Assessment and Measures Development by Neely [6], cited by Bourne [21], also cited as Cambridge Performance Measurement Process by Pun and White [11].
- Workflow-based monitoring, Self-Assessment, Benchmarking and Statistical Process Control, amended in Kueng and Krahn [23].

The only PMSs that could be identified as not exclusively using numeric indicators were

- EFQM by Moll [45], discussed by Kueng and Krahn [23], Beatham [35], Lönnqvist [9], Craciun and Criveanu [36], Folan and Browne [37] as well as Pun and White [11].
- Process Performance Measurement System, introduced by Kueng and Krahn [23].

Alternatively, they consider *qualitative* aspects of performance through the definition of descriptive goals and recommend the use of secondary scores or indicators that state to what extent the goals are fulfilled.

E. Findings

As a first result of the investigation, principal properties and limitations of PMS actually can be collected and *clustered* to specific viewpoints.

One of the main restrictions to these PMS is that business processes that cannot be measured easily are likely to be either *left out* or subject to costly design of artificial measures that suit in the numeric framework. Closely related

to the collection of constraints, the design of PMS in the present investigation is still predominantly driven by numeric parameters; seventeen out of nineteen reviewed PMS rely on measures or KPIs.

Testing the evaluated properties and limitations of PMS against the four introductory commands of transparency, comparability and domain independence of business processes as well as visibility of process performance, following findings can be concluded:

A definitely positive aspect to *transparency* is that the mere existence of a PMS in an enterprise drives towards improved process definition and documentation. Transparency would even increase if the measures would be correctly linked back to department goals, strategy, financial aspects or culture, like demanded by many reviewers. The same is true for the demands of better *accessibility* of measures and results as well as for systematic pruning of obsolete measures. Both points are affecting a better *comparability*, too.

Given a PMS that is self-contained in an enterprise ecosystem, its measures, results and hence business processes can of course be compared to each other and a change over time and between different objects of measurement will be visible in case their definitions do neither change significantly in the rendered time or scope nor differ much between departments or divisions. *Comparability* may also increase if structured review loops are performed to ensure consistence of measures, or not too many measures are in place that could dilute the overall picture and impression.

Moreover, most PMS are able to support the assessment of the entire business performance and are as such not restricted to specific *functional domains* unless they intentionally differentiate between local and global optimization. This contradicts the historical isolated and ad-hoc usage of KPIs without PMS frameworks, e.g. in production or finance.

Only two of the 19 investigated PMS use others than numeric parameters, in this case goals to describe and assess process performance. So they also support the idea of extending the indicator ecosystem to non-numeric measures. Hence, they support the concept of *visibility* of process performance through an *additional* viewpoint which per definition increases visibility, at least in the context of use of exactly these PMS.

Even more problems arise of the fact that many designers and users of PMS simply *ignore* or avoid to aggregate and redesign difficult qualitative or hard-measurable measures so that they fit in numeric systems.

On the one hand, *transparency* lacks because this behavior virtually *excludes* processes and assessment aspects in principle and for any assessment, they are simply not existent. On the other hand, *comparability* will be bad when measures are artificially built or simplified. Especially when using such measures, the *consistence* of definitions over time and scope has to be ensured because the sources of the definition, way of calculation, measurement or results might

not be self-explaining or sustained. Finally, *visibility* remains restricted as is was if the PMS of choice relies on only one indicator, namely KPIs.

This means: To use a conventional PMS adds up transparency and comparability to business processes (and this enhances business *performance* when problems can be seen and improvements can be put in place), even more, if the used PMS is built in a way to overcome the restrictions that can be clustered to the three viewpoints. But it would be even better to use additional indicators because through better *visibility*, the perception of previously excluded or invisible problems would reveal *additional* improvement possibilities.

IV. CONCLUSION AND OUTLOOK

According to the investigation findings, the problems when using conventional PMS can be examined from three viewpoints and tested on their effect on transparency, comparability, domain independence of processes and visibility of performance.

In particular, some constraints in process performance remain *invisible* when using numeric parameters/KPIs for assessment exclusively, which is the predominant method for the PMS taken into account. Of course the use of KPIs might be *sufficient* for a lot of performance problems, as broad application in the past proved. But at least for enterprises that experience problems with PMS restricted to numeric parameters, there is a necessity for a concept of a PMS (framework) that also integrates assessment methods beyond numeric indicators. It can be postulated for two reasons:

- (1) For reasons of better visibility, comparability and transparency it should contain more suitable approaches that describe and assess the performance of a process rather than putting a distinctive numeric value on it. So, improvements in process efficiency and effectiveness that contribute to the enterprises' success can *better* be visualized. It would even address process problems that have been invisible to KPIs before, but are accessible to other *indirect* indicators like success factors, soft goals, milestones, complexity, relations or maturity.
- (2) For reasons of business resource awareness, it would allow enterprises to directly assess remaining hard measurable business processes and performance problems *without* extra cost of finding and applying above mentioned additional models or defining and maintaining stopgap numeric measures. Moreover, previously *invisible* value-adding additional insights for performance problems are most likely through using uncommon approaches to business process assessment.

So when business performance *increases* by the number of *improvements*, which are more likely to be established in *transparent* and *comparable* business processes with *visible* performance, there is a certain danger of *not* revealing problems and not implementing improvements with

conventional KPI-using PMS - and hence, to *lose* money by *wasting* business performance.

All in all, a need for an integrated *performance assessment system* beyond numeric indicators using additional *indirect indicators* can be postulated. Deployment and the description of possible contents will be subject to further research based on these and prior findings.

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